

COMPETENCY-BASED LEARNING MATERIALS (CBLM)

AI BASIC PYTHON DEVELOPER

TECHNICAL AND VOCATIONAL CERTIFICATE I — LEVEL II

Core Competency: Apply Structured Computer Programming

Module Title: Apply Structured Computer Programming

Learning Outcome: LO1 – Setup Python on Windows and Other Operating Systems

Nominal Duration:

LEARNING OUTCOME 01

Setup Python on Windows and Other Operating Systems

Upon completion of this Learning Outcome, students/trainees will have the following competencies:

1. Explain the basic concepts and uses of Python.
2. Explain basic Python syntax and its characteristics.
3. Identify the key differences between Python 2 and Python 3.
4. Explain the purpose of environment variables and system PATH.
5. Explain the difference between Python Shell and IDLE.
6. Explain compiler and interpreter concepts.
7. Identify Python IDEs and code editors.
8. Select an appropriate Python IDE or code editor.
9. Download Python installation files from official sources.
10. Install the Python interpreter on Windows.
11. Configure Python environment variables and system PATH settings when required.
12. Install and configure Python IDLE or another appropriate Python development environment.
13. Verify Python installation using command-line tools.
14. Execute Python commands using interpreter mode.
15. Create and execute Python script files using script mode.

INFORMATION SHEET No. 1: Introduction to Python and Python Development Environment

Objective

Upon reading this Information Sheet, the trainee must be able to:

- Explain what Python is.
- Explain why Python is used and what Python can do.
- Describe basic Python syntax.
- Compare Python syntax with other programming languages.
- Identify the key differences between Python 2 and Python 3.
- Explain environment variables and the PATH setting.
- Explain Python Shell and IDLE.
- Distinguish a compiler from an interpreter.
- Identify Python IDEs and code editors.
- Select an appropriate Python development environment.
- Describe the basic process of installing Python.

1. What is Python?

Python is a high-level, general-purpose programming language used to develop different types of computer programs and applications.

Python was designed with an emphasis on readability and relatively simple syntax. It is suitable for beginners and is also widely used by professional developers.

Python can be used for:

- Basic programming
- Software development
- Web development
- Data processing
- Data analysis
- Artificial Intelligence
- Machine Learning
- Automation
- Scientific computing
- Network programming
- Testing
- Scripting

For an **AI Basic Python Developer**, Python provides the programming foundation needed for later learning about data, artificial intelligence and machine learning.

2. Why Python?

Python is widely used because it provides several advantages.

2.1 Readable Syntax

Python syntax is designed to be relatively easy to read.

Example:

```
print("Hello Python")
```

The program displays:

```
Hello Python
```

2.2 Easy to Learn

Python uses a relatively small number of basic programming structures, making it suitable for beginning programmers.

2.3 Cross-Platform

Python can be installed and used on different operating systems, including:

- Windows
- macOS
- Linux

2.4 Large Library Ecosystem

Python provides many libraries and packages.

Examples include:

- NumPy
- Pandas
- Matplotlib
- Scikit-learn
- TensorFlow

2.5 Artificial Intelligence

Python is widely used for:

- Machine Learning
- Deep Learning
- Computer Vision
- Natural Language Processing
- Data Science

3. What Can Python Do?

Python can be used to develop many different types of programs.

Examples

Simple calculation

```
total = 10 + 20  
print(total)
```

Text processing

```
name = "Python"  
print(name)
```

File processing

Python can read and write files.

Data analysis

Python libraries such as Pandas can process datasets.

Artificial Intelligence

Python can be used with machine-learning libraries to create AI applications.

Automation

Python can automate repetitive computer tasks.

4. Python Syntax

Syntax means the rules used to write instructions in a programming language.

Example:

```
name = "Dara"  
age = 20  
  
print(name)  
print(age)
```

Python uses indentation to identify blocks of code.

Example:

```
age = 20  
  
if age >= 18:  
    print("Adult")
```

The spaces before print() are important because they indicate that the statement belongs to the if block.

5. Python Syntax Compared with Other Programming Languages

Different programming languages use different syntax rules.

For example, Python uses indentation to define code blocks:

```
if age >= 18:  
    print("Adult")
```

A language such as C# commonly uses braces:

```
if (age >= 18)  
{  
    Console.WriteLine("Adult");  
}
```

Python does not normally require semicolons at the end of every statement.

Python:

```
name = "Dara"  
age = 20
```

Some other programming languages commonly use semicolons:

```
name = "Dara";  
age = 20;
```

The important point is that each programming language has its own syntax rules.

6. Python 2 and Python 3

Python 2 and Python 3 are major versions of Python.

Python 2 is an older version and is no longer recommended for new development.

This training program uses **Python 3.x**.

One visible difference is the print statement/function.

Python 3:

```
print("Hello")
```

Python 2 commonly used:

```
print "Hello"
```

Another important difference is that Python 3 treats text and binary data differently from Python 2.

For this reason, trainees must use Python 3 for this training program.

7. Environment Variables

An environment variable is a system setting containing information that can be used by the operating system or applications.

One important environment variable is:

PATH

PATH tells the operating system where it can search for executable programs.

If Python is correctly configured in PATH, a user may execute:

```
python --version
```

from Command Prompt or Terminal.

The result should display the installed Python version.

8. Python Installation and PATH

During Python installation, the installer may provide an option to make Python available through PATH.

When the option is appropriate for the training environment, the trainee should select it according to the installation instructions.

After installation, Python can be tested using:

```
python --version
```

or, on Windows systems where the Python launcher is available:

```
py --version
```

If Python cannot be found, the PATH configuration may need to be checked.

Trainees should not modify unrelated system environment variables.

9. Python Shell

The Python Shell is an interactive environment where Python commands can be entered and executed immediately.

Example:

```
>>> 10 + 20
30
```

Another example:

```
>>> print("Hello")
Hello
```

The Python Shell is useful for:

- Learning Python
- Testing commands
- Testing expressions
- Performing calculations
- Checking small pieces of code

10. Python IDLE

IDLE stands for: **Integrated Development and Learning Environment**

IDLE is a simple development environment that normally comes with the standard Python installation.

It provides:

- Python Shell
- Code editor
- Syntax highlighting
- Program execution
- Basic debugging support

IDLE is particularly useful for beginners learning Python.

11. Python Shell vs IDLE

The Python Shell is an interactive execution environment.

IDLE is a development environment that provides both a Python Shell and an editor.

Python Shell	IDLE
Mainly used for interactive commands	Provides Shell and Editor
Commands can be executed immediately	Programs can be created and saved
Useful for quick testing	Useful for learning and developing programs
Can be accessed through Terminal/Command Prompt	Provides a graphical development environment

12. Interpreter

An interpreter is software that executes programming instructions through an interpreter environment.

Python programs are commonly executed using the Python interpreter.

For example:

```
print("Python")
```

The Python interpreter processes the instruction and produces:

Python

13. Compiler vs Interpreter

A compiler generally translates source code into another executable or intermediate form before the program is run.

An interpreter executes program instructions through an interpreter environment.

At this Level II stage, trainees should understand the basic distinction:

Compiler	Interpreter
Generally translates source code before execution	Executes instructions through an interpreter
Often produces compiled output	Usually does not require a separate executable for each script
Compilation is generally a separate stage	Execution can occur through the interpreter
Examples include C/C++ compilation tools	Python is commonly taught as an interpreted language

Python implementations also use intermediate bytecode and a virtual machine internally, so the simple “compiler versus interpreter” distinction is an introductory model rather than a complete description of Python implementation.

14. Script Mode

A Python script is a file containing Python source code.

Python source files normally use:

.py

Example:

hello.py

The file can contain:

```
print("Hello Python")  
print("AI Basic Python Developer")
```

The program can then be saved and executed.

15. Interpreter Mode vs Script Mode

Interpreter/Shell Mode	Script Mode
Commands are entered directly	Code is stored in a file
Results are displayed immediately	Complete program can be executed
Good for testing	Good for program development
Code may not be saved automatically	Source code can be saved
Uses Python prompt	Uses .py files

The trainee must be able to identify and use both modes.

16. Python IDEs and Code Editors

An IDE is an Integrated Development Environment.

A code editor is software designed primarily for editing source code, often with additional programming features.

Examples:

IDLE:

Simple Python development environment suitable for beginners.

Visual Studio Code:

A lightweight and extensible code editor that can be configured for Python development.

PyCharm:

A Python-focused integrated development environment.

Jupyter Notebook:

An interactive environment widely used for data analysis, scientific computing and AI learning.

Visual Studio:

A full-featured development environment that can support Python development through appropriate components.

17. How to Choose a Python IDE

The trainee should consider:

1. Level of programming experience
2. Type of project
3. Operating system

4. Required features
5. Ease of installation
6. Debugging capability
7. Extension/package support
8. Available computer resources
9. Training requirements
10. Organizational requirements

For beginners in this Level II program:

- **IDLE:** is appropriate for basic Python learning.
- **Visual Studio Code:** is appropriate when a more flexible development environment is required.

18. Installing Python on Windows

The general procedure is:

PYTHON INSTALLATION & EXECUTION WORKFLOW

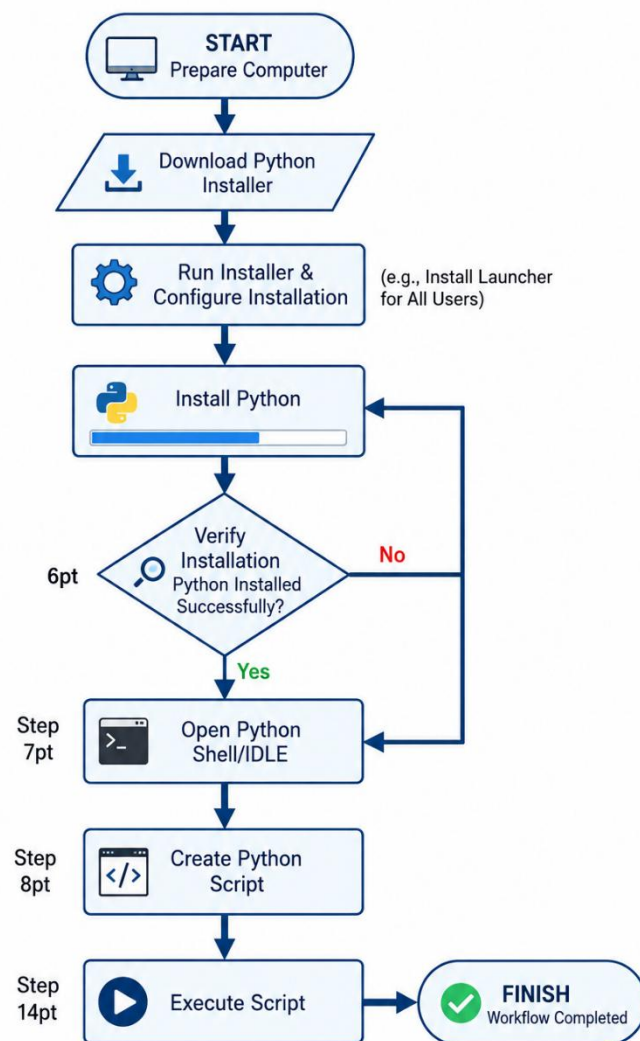


Figure 1: Python Installation workflow

The trainee must follow the installation instructions provided by the trainer and the software installer.

19. Python on Other Operating Systems

Python can also be installed on:

Windows

Python installer can be used to install Python.

macOS

Python can be installed using an appropriate installer or package-management method according to the training environment.

Linux

Python is commonly available through the operating system's package-management system.

The exact installation procedure can differ between operating systems.

The trainee should always follow the appropriate installation procedure for the operating system being used.

20. Basic Installation Verification

After installation, open Command Prompt or Terminal and execute:

```
python --version
```

If available, Windows users may also test:

```
py --version
```

Then start Python:

```
python
```

Test:

```
print("Python installation successful")
```

Expected output:

```
Python installation successful
```

```
Exit:
```

```
exit()
```

SELF-CHECK No. 1

Instructions

Answer all questions. Do not refer to the Information Sheet while answering.

Part A: Multiple Choice

1. What is Python?

- A. An operating system
- B. A high-level programming language
- C. A database
- D. A web browser

2. Which version of Python is used for this training?

- A. Python 1
- B. Python 2
- C. Python 3
- D. Python 0

3. Which file extension is normally used for Python source files?

- A. .cpp
- B. .java
- C. .py
- D. .exe

4. Which environment variable is commonly used to locate executable programs?

- A. TEMP
- B. PATH
- C. USER
- D. DATA

5. What does IDLE stand for?

- A. Integrated Development and Learning Environment
- B. Internet Development Learning Environment
- C. Integrated Data Language Editor
- D. Internet Development Logic Engine

6. Which mode allows Python commands to be executed immediately?

- A. Script mode
- B. Interpreter/Shell mode
- C. Compile mode
- D. Design mode

7. Which command can be used to check the installed Python version?

- A. python -version
- B. python install
- C. python check
- D. python setup

8. Which of the following is a Python development environment?

- A. IDLE
- B. Calculator
- C. Paint
- D. File Explorer

9. Which Python version is recommended for new development in this course?

- A. Python 1
- B. Python 2
- C. Python 3
- D. Python Classic

10. Which of the following is an application of Python?

- A. Artificial Intelligence
- B. Only typing documents
- C. Only printing
- D. Only playing music

ANSWER KEY No. 1

Part A

No.	Answer
1	B
2	C
3	C
4	B
5	A
6	B
7	A
8	A
9	C
10	A

OPERATION SHEET No. 1:

Title: Install and Configure Python Development Environment

Objective

Upon completion of this Operation Sheet, the trainee must be able to:

- Prepare a computer for Python installation.
- Download the Python installer.
- Run the Python installer.
- Complete the installation process.
- Configure PATH when required.
- Verify the Python installation.
- Open Python Shell.
- Open Python IDLE.

Tools, Equipment and Materials

No.	Description	Unit	Qty
1	Desktop Computer/Laptop	Unit	1
2	Windows Operating System	Unit	1
3	Internet Connection	Set	1
4	Python 3 Installer	Copy	1
5	Python Installation Guide	Copy	1
6	CBLM	Copy	1
7	USB Flash Drive	Unit	1

Procedure/Step | Key Performance

No.	Procedure/Step	Key Performance
1	Prepare the computer	Computer starts and operates normally
2	Connect to the Internet	Internet connection is available
3	Open a web browser	Browser opens successfully
4	Obtain the Python installer	Appropriate Python 3 installer is selected
5	Save the installer	Installer is saved in the designated location
6	Run the installer	Installer starts correctly
7	Read installation options	Options are understood before proceeding
8	Configure PATH when required	Appropriate PATH option is selected
9	Start installation	Installation begins correctly
10	Complete installation	Python installation finishes without error
11	Open Command Prompt	Command Prompt opens correctly
12	Verify Python	<code>python --version</code> displays Python version
13	Start Python	Python interpreter starts
14	Execute a command	Python command produces correct result
15	Open IDLE	IDLE opens successfully
16	Test IDLE	Python command executes correctly

Detailed Operation Procedure

Step 1: Prepare the computer

1. Turn on the computer.
2. Log in using the authorized account.
3. Check the operating system.
4. Confirm sufficient storage space.
5. Connect to the Internet.

Step 2: Obtain the Python installer

1. Open a web browser.
2. Go to the approved official Python source.
3. Select the appropriate Python 3 installer.
4. Download the installer.
5. Save it in the assigned folder.

Step 3: Run the installer

1. Locate the downloaded installer.
2. Double-click the installer.
3. Read the installation options.
4. Follow the installation instructions.

Step 4: Configure PATH

If the installer provides an option to add Python to PATH and this is required by the training environment, select the appropriate option.

The trainee must not change unrelated environment variables.

Step 5: Complete installation

1. Start installation.
2. Wait for the installation process to complete.
3. Confirm that the installation finishes successfully.
4. Close the installer.

Step 6: Verify Python

Open Command Prompt and type:

```
python -version
```

Record the result:

Python Version: _____

Step 7: Start Python interpreter

Type:

```
python
```

Confirm that the Python prompt appears.

Step 8: Execute a Python command

Type:

```
print("Python is installed successfully")
```

Expected result:

Python is installed successfully

Step 9: Exit Python

Type:

```
exit()
```

Step 10: Open IDLE

1. Open the Start menu.
2. Search for IDLE.
3. Open Python IDLE.
4. Confirm that the Python Shell appears.
5. Execute:

```
print("IDLE is working")
```

Step 11: Record the result

IDLE Result: _____

Safety Precautions

1. Use an authorized computer.
2. Download software only from trusted/approved sources.
3. Do not open unknown executable files.
4. Do not modify unrelated system settings.
5. Protect user and institutional information.
6. Report errors to the trainer.
7. Keep installation files organized.
8. Follow laboratory rules.

TASK SHEET No. 1.1

Title: Verify Python Interpreter and Python Shell

Performance Task:

The trainee shall verify the installed Python interpreter and practice executing Python commands using Python Shell/interpreter mode.

Performance Objective:

Given a computer with Python installed, Command Prompt or Terminal, and the required CBLM, the trainee must be able to verify Python installation and execute Python commands correctly.

Description:

The trainee will perform individual practice activities to:

1. Check the Python version.
2. Start Python interpreter mode.
3. Execute Python expressions.
4. Create variables.
5. Display information.
6. Exit Python interpreter mode.

Picture/Overview:

PYTHON COMMAND PROMPT & SHELL WORKFLOW



Figure 2: Python Command Prompt Workflow

Tools, Equipment and Materials:

No.	Description	Unit	Qty
1	Desktop Computer/Laptop	Unit	1
2	Windows Operating System	Unit	1
3	Python 3	Copy	1
4	Command Prompt/Terminal	Software	1
5	CBLM	Copy	1
6	Worksheet	Copy	1

Supplier:

- Training institution

- Authorized software provider
- Official Python software source

Procedure/Step | Key Performance

No.	Procedure/Step	Key Performance
1	Open Command Prompt	Command Prompt opens correctly
2	Check Python version	Python version is displayed
3	Start Python	Python interpreter starts
4	Check Python prompt	>>> prompt is displayed
5	Execute calculation	Correct result is displayed
6	Create variable	Variable is created correctly
7	Display variable	Correct value is displayed
8	Execute print command	Correct message is displayed
9	Exit Python	Interpreter closes correctly

Practical Activities

Activity 1: Check Python Version

Execute:

```
python --version
```

Record:

Version: _____

Activity 2: Start Python

Execute:

```
python
```

Confirm that the Python prompt appears.

Activity 3: Perform Calculation

Execute:

```
10 + 20
```

Expected result:

```
30
```

Activity 4: Create Variables

Execute:

```
name = "Python Student"
```

```
age = 20
```

Then:

```
print(name)
```

```
print(age)
```

Activity 5: Display a Message

Execute:

```
print("I am learning Python programming")
```

Activity 6: Exit Python

Execute:

```
exit()
```

Assessment Method:

- Direct observation
- Practical demonstration
- Oral questioning
- Performance checklist

TASK SHEET No. 1.2

Title: Create and Execute Python Script Using IDLE

Performance Task:

The trainee shall create, save and execute a Python script using Python IDLE.

Performance Objective:

Given a computer with Python IDLE installed, the trainee must create and execute a Python .py file correctly.

Description:

The trainee will practice script mode by creating a simple student information program.

Picture/Overview:

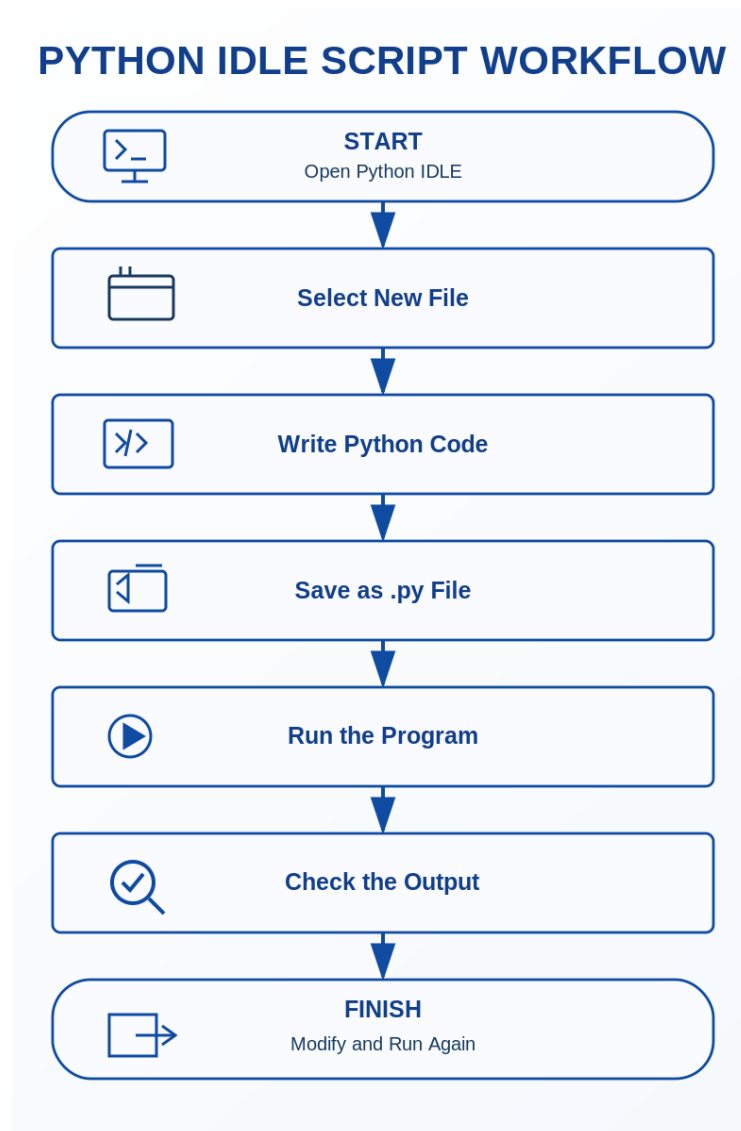


Figure 3: Python IDLE

Tools, Equipment and Materials:

No.	Description	Unit	Qty
1	Desktop Computer/Laptop	Unit	1
2	Windows Operating System	Unit	1
3	Python 3	Copy	1
4	Python IDLE	Software	1
5	CBLM	Copy	1
6	Worksheet	Copy	1

Supplier:

- Training institution
- Authorized software provider
- Official Python software source

Procedure/Step | Key Performance

No.	Procedure/Step	Key Performance
1	Open IDLE	IDLE opens successfully
2	Create a new file	New editor window opens
3	Enter Python code	Correct syntax is entered
4	Save the file	File is saved with .py extension
5	Run the program	Program executes successfully
6	Check output	Expected output is displayed
7	Modify the program	Required information is changed
8	Run again	Modified output is displayed

Practical Activity

Open IDLE and create a new file.

Save it as:

student_information.py

Enter:

```
name = "Your Name"
age = 20
course = "AI Basic Python Developer"
level = "Level II"

print("Student Information")
print("-----")
print("Name:", name)
print("Age:", age)
print("Course:", course)
print("Level:", level)
```

Run the program.

Then change the name and age values and run the program again.

Expected Result:

```
Student Information
-----
Name: Your Name
Age: 20
Course: AI Basic Python Developer
Level: Level II
```

Assessment Method:

- Direct observation
- Practical demonstration
- Review of .py file
- Output verification
- Performance checklist

TASK SHEET No. 1.3

Title: Select and Configure a Python IDE or Code Editor

Performance Task:

The trainee shall select and configure an appropriate Python development environment for basic Python programming.

Performance Objective:

Given a computer and available Python development environments, the trainee must select an appropriate environment and demonstrate that it can create and execute a Python program.

Description:

The trainee will compare available Python development environments and select one suitable for Level II Python programming.

Possible environments include:

- IDLE
- Visual Studio Code
- PyCharm
- Jupyter Notebook
- Other approved Python development environments

Picture/Overview:



Figure 4: Python IDE Selection Flow

Tools, Equipment and Materials:

No.	Description	Unit	Qty
1	Desktop Computer/Laptop	Unit	1
2	Python 3	Copy	1
3	Python IDE/Code Editor	Software	1
4	Internet Connection	Set	1
5	CBLM	Copy	1
6	Worksheet	Copy	1

Supplier:

- Training institution
- Authorized software provider
- Official software source

Procedure/Step | Key Performance

No.	Procedure/Step	Key Performance
1	Identify programming requirements	Requirements are correctly identified
2	Identify available IDEs	Appropriate options are identified
3	Compare IDE features	Features are correctly compared
4	Select an IDE	Appropriate IDE is selected
5	Configure Python support	Python support operates correctly
6	Create a Python file	.py file is created
7	Execute the program	Program executes successfully
8	Verify output	Correct output is displayed

Practical Activity

Complete the following table:

IDE/Code Editor	Main Feature	Suitable for Beginner?	Selected
IDLE			
Visual Studio Code			
PyCharm			
Jupyter Notebook			

Select one environment and explain why it is appropriate for your Python training.

Selected IDE: _____

Reason:

Assessment Method:

- Observation
- Oral questioning
- Practical demonstration
- Completed comparison table
- Performance checklist

JOB SHEET No. 1

Title: Prepare and Test a Python Development Environment

Performance Task:

The trainee shall independently install, configure and test a Python development environment for basic Python programming.

Performance Objective:

Given a computer, operating system, Internet connection, Python installation file, Python IDLE or approved IDE, and training materials, the trainee must prepare a working Python development environment and demonstrate interpreter and script modes.

Description:

You have been assigned a computer that will be used for Python programming training.

Your task is to prepare the computer for use by:

1. Obtaining Python.
2. Installing Python.
3. Configuring PATH when required.
4. Installing or opening Python IDLE.
5. Verifying Python.
6. Executing Python interpreter commands.
7. Creating a Python script.
8. Executing the script.
9. Demonstrating the difference between interpreter mode and script mode.

Picture/Overview:



Figure 5: Job Sheet Flow

Tools, Equipment and Materials:

No.	Description	Unit	Qty
1	Desktop Computer/Laptop	Unit	1
2	Windows Operating System	Unit	1
3	Internet Connection	Set	1
4	Python 3 Installer	Copy	1
5	Python IDLE/Approved IDE	Software	1
6	CBLM	Copy	1
7	Installation Guide	Copy	1
8	USB Flash Drive	Unit	1

Supplier:

- Training institution
- Authorized software provider
- Official Python software source

Procedure/Step | Key Performance

No.	Procedure/Step	Key Performance
1	Prepare workstation	Computer is ready for installation
2	Obtain Python installer	Correct Python 3 installer is downloaded
3	Run installer	Installer runs correctly
4	Follow installation instructions	Instructions are followed correctly
5	Configure PATH	PATH is configured when required
6	Complete installation	Python installs successfully
7	Open Command Prompt	Command Prompt operates correctly
8	Verify Python	Version is displayed
9	Start interpreter	Python Shell opens
10	Execute command	Correct result is produced
11	1Open IDLE	IDLE opens successfully
12	1Create script	.py file is created
13	Save script	File is saved correctly
14	Execute script	Script runs successfully
15	Demonstrate interpreter mode	Trainee demonstrates interactive execution
16	Demonstrate script mode	Trainee demonstrates .py file execution
17	Organize files	Files are stored appropriately
18	Apply safety practices	Safe and responsible procedures are followed

Job Program

Create a file named:

`python_environment_test.py`

Enter:

```
print("=====")
print("      PYTHON ENVIRONMENT TEST")
print("=====")
print("Course : AI Basic Python Developer")
print("Level  : II")
print("Status : Python environment is ready")
```

Run the program.

Expected Output:

```
=====
      PYTHON ENVIRONMENT TEST
=====
Course : AI Basic Python Developer
Level  : II
Status : Python environment is ready
```

Assessment Method:

- Direct observation
- Practical demonstration
- Product/output inspection
- Oral questioning
- Performance checklist

PERFORMANCE CHECKLIST No. 11

Task: Prepare and Test a Python Development Environment

Name: _____

Date: _____

Did you use all these performance criteria during your task?

No.	Criteria	Yes	No
1	Downloaded the Python installer from an official/trusted source	☐	☐
2	Selected the appropriate Python version	☐	☐
3	Ran the Python installer correctly	☐	☐
4	Followed the installation instructions	☐	☐
5	Configured the Python environment appropriately	☐	☐
6	Configured PATH when required	☐	☐
7	Completed the Python installation successfully	☐	☐
8	Downloaded/located Python IDLE appropriately	☐	☐
9	Completed the IDLE installation/setup process as instructed	☐	☐
10	Verified Python installation using a command-line tool	☐	☐
11	Executed Python commands using interpreter mode	☐	☐
12	Created a Python script file using .py extension	☐	☐
13	Executed the Python script using script mode	☐	☐
14	Correctly identified interpreter mode	☐	☐
15	Correctly identified script mode	☐	☐
16	Distinguished Python Shell from IDLE	☐	☐
17	Selected an appropriate Python IDE/code editor	☐	☐
18	Followed software and computer safety procedures	☐	☐
19	Demonstrated quality consciousness and attention to detail	☐	☐
20	Demonstrated patience, diligence and responsibility	☐	☐
21	Maintained organized files and workstation	☐	☐
22	Completed the assigned job within the allocated time	☐	☐

Comment:

Overall Assessment

Competent: ☐ Yes ☐ No

Needs Further Practice: ☐ Yes ☐ No

Trainer's Name: _____

Trainer's Signature: _____

Date: _____